



Impacts of Urbanization on Hummingbird Diet & Coloration



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Background

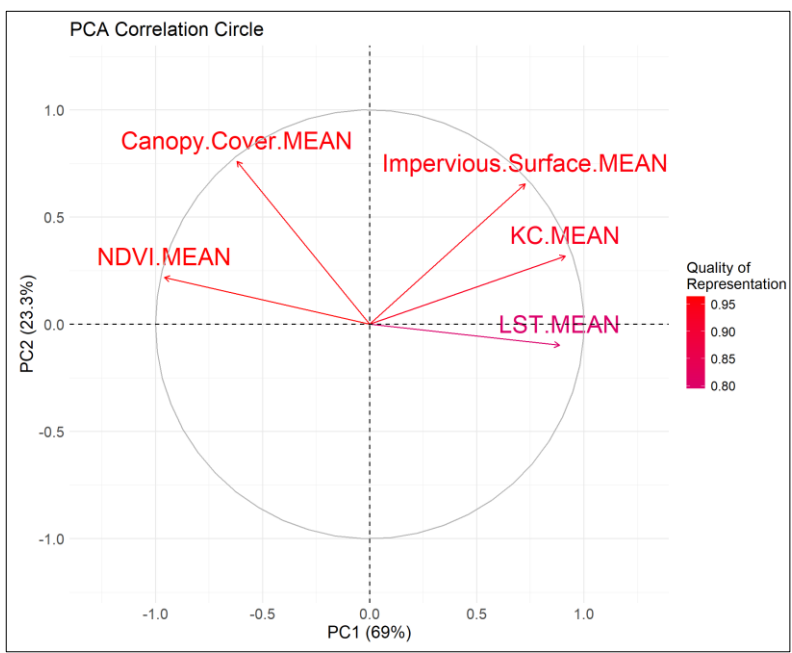
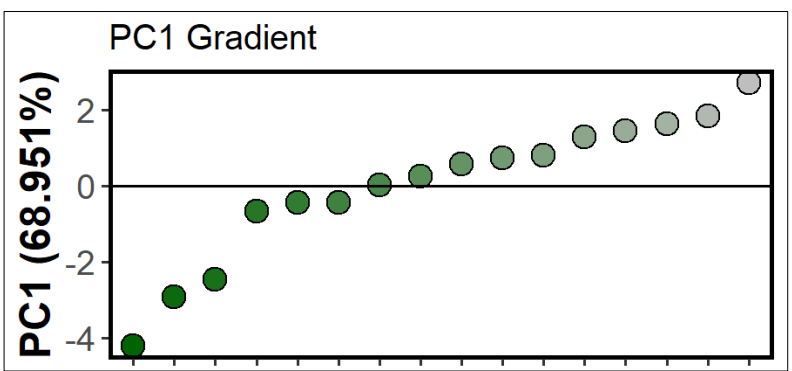
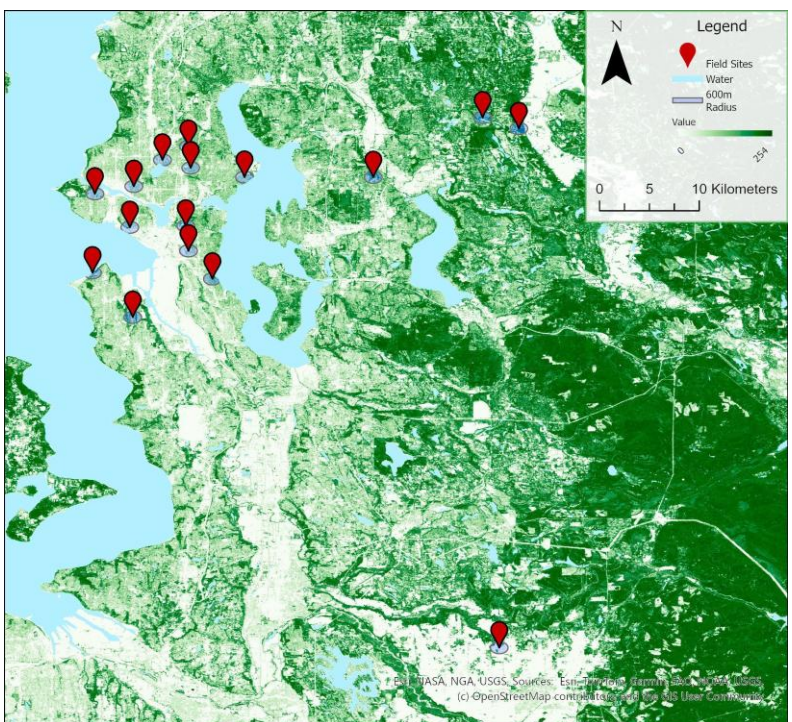
- Anna's Hummingbirds (*Calypte anna*) wintering ranges are expanding northwards, likely due to access to feeders, climate change, and urban heat islands¹.
- Experimentally increasing the protein in male *C. anna* diets has been shown to increase red chroma and shift towards longer, more red wavelengths in crown feathers².
- Urbanization has significantly decreased arthropod abundance and richness in urban areas, which may drive dietary differences in *C. anna* across an urban-to-rural gradient³.
- Stable isotope analyses can inform us about hummingbird diets. For example, higher N15 can be detected in feathers of hummingbirds during seasons in which they are known to consume more arthropods⁴. Differences in C13:12 ratios have been detected in the breath of hummingbirds feeding on artificial nectar (mostly sugar cane) vs. natural nectar, though no research has been conducted looking at whether this signal is detectable in feathers.
- **Hypothesis:** If *C. anna* in more urbanized environments have reduced access to arthropod food resources, we expect to see differences in their coloration and stable isotope ratios. Specifically, we predict that urban birds will have a decrease in brightness, chroma, and red chroma, and a shift away from a redder hue. We also hypothesized a decrease in the ratio of nitrogen 15:14, and an increase in the carbon 13:12 ratio due to increased use of feeders.



Credit: Chris Steinbronn

Methods

- *C. anna* (n=27) were captured using drop traps at artificial feeders in Seattle, Washington during the winter (Jan-Feb, n=14) and summer (Jun-Jul, n= 13), 2026. One tail (R4), 3 gorget, and 3 crown feathers were collected from each bird.
- A field site's urbanization score was determined by a principle component analysis of 5 remote sensing variables: Landsat land surface temperature, Landsat NDVI (Normalized Difference Vegetation Index), impervious surface cover, air temperature, and canopy cover.
- The ratio of nitrogen 15:14 ($\delta^{15}\text{N}$) and carbon 13:12 ($\delta^{13}\text{C}$) were analyzed from tail feathers at a private stable isotope lab, with $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ of the feathers being compared to Air N2 and VPDB, respectively.
- Colorimetric variables were measured using an Ocean Optics Miniature spectrometer and DH-2000-BAL light source. Reflectance was measured at multiple angles to identify maximum reflectance of each feather/color patch. The colorimetric variables were calculated using Pavo (R package).



Results

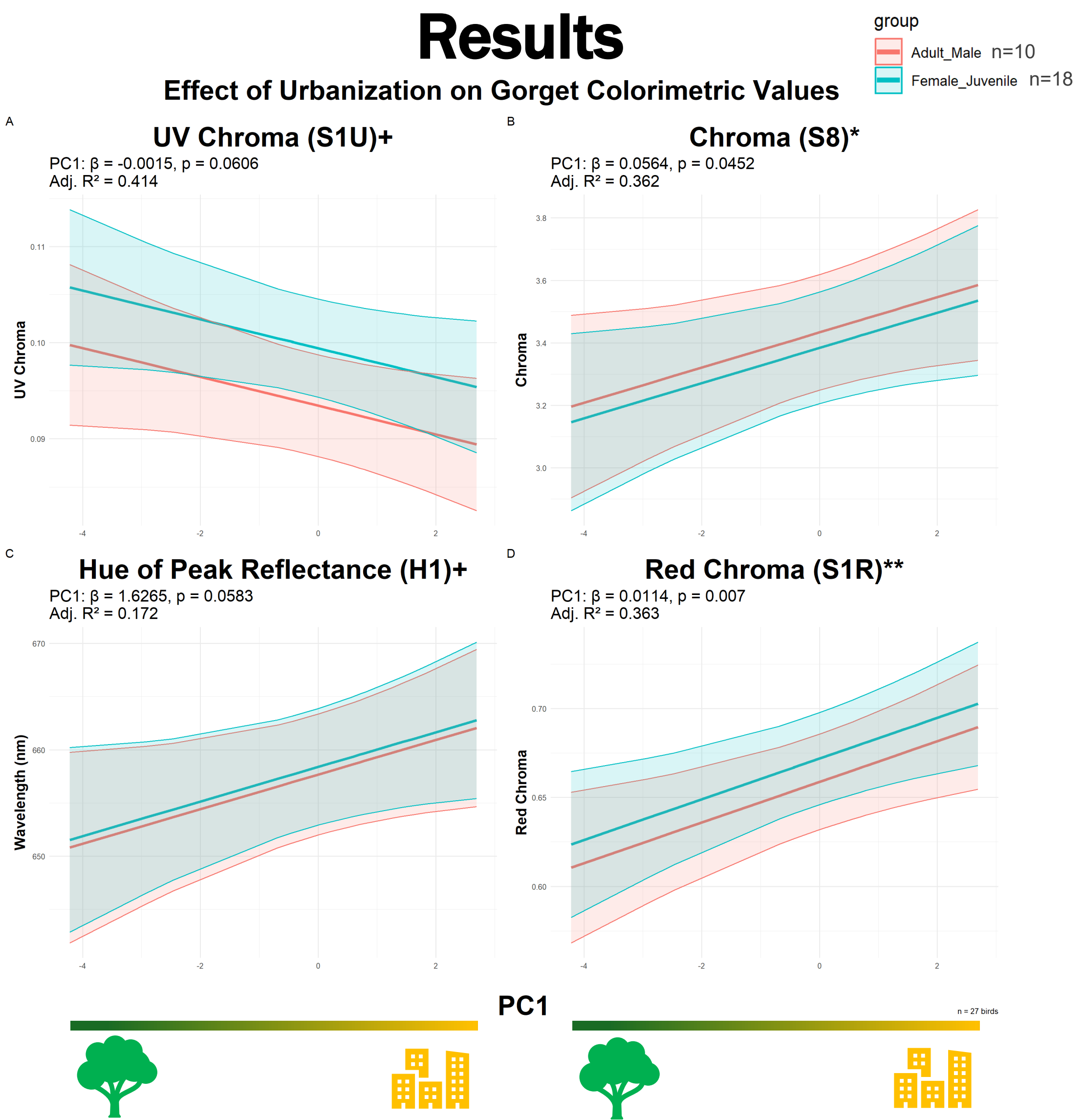


Figure 1. Predicted relationships between urbanization (PC1) and gorget color metrics. (A) UV Chroma exhibited a negative relationship with urbanization. (B) Chroma increased significantly with increasing urbanization. (C) Hue tended to shift toward longer (redder) wavelengths with increasing urbanization. (D) Red chroma increased significantly with increasing urbanization. Shaded ribbons represent 95% confidence intervals. Models included body mass, sex-age class, and time of capture (season) as covariates. Season: S1U+, S8**, H1*, S1R** + p < 0.10, * p < 0.05, ** p < 0.01.

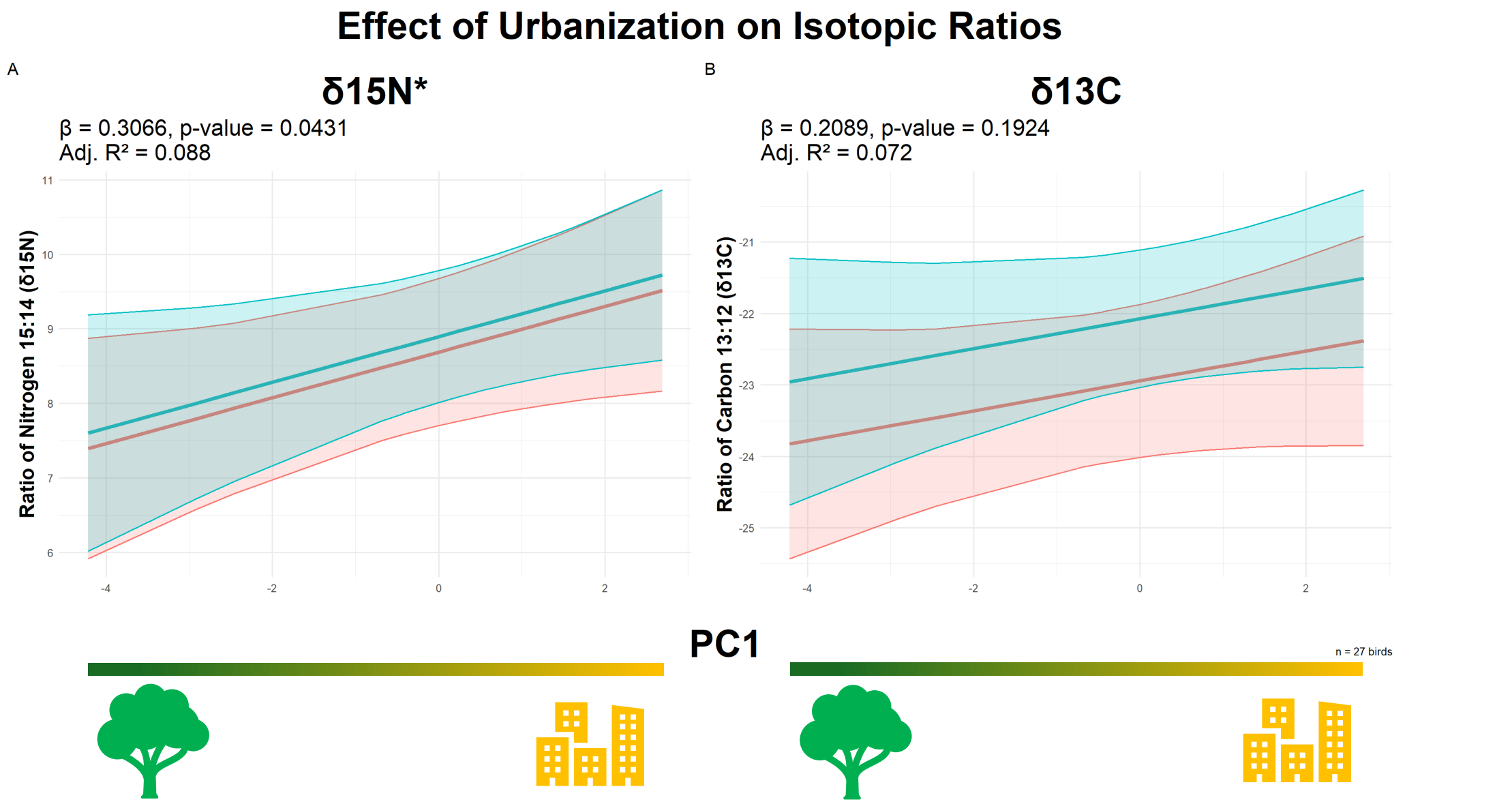


Figure 2. Predicted relationships between urbanization (PC1) and stable isotope ratios. (A) $\delta^{15}\text{N}$ increased significantly with urbanization (B) $\delta^{13}\text{C}$ has a positive, but insignificant relationship with urbanization. Shaded ribbons represent 95% confidence intervals. All models included sex-age class, and time of capture (season) as covariates. There was no significant effect of season. + p < 0.10, * p < 0.05, ** p < 0.01.

Discussion

- **We found:**
 - A significant increase in overall chroma and red chroma, and trends towards redder hues and decreased UV chroma as urbanization increased.
 - $\delta^{15}\text{N}$ increased significantly as urbanization increased. We are the first to demonstrate that variation in $\delta^{13}\text{C}$ can be detected in the feathers of hummingbirds and may reflect artificial feeder use though trend was not significant.
- While one study has shown a positive association between protein intake and red chroma/hue, we did not find a significant relationship between diet and coloration. Our findings suggest that the color shifts observed here may be driven by other variables associated with urbanization, other than protein intake.
- Ecological factors such as sexual selection, changes in signal to noise, genetic drift, or selection for hues that are less visible to novel predators (e.g. cats) could be responsible for the shifts along the urban gradient.
- **Future directions:** We will use meta-barcoding to characterize diet from fecal samples collected from these same individuals



Acknowledgments: Chris Steinbronn for the photographs. IEH analytical laboratories: Sheel Prajapati and Raquel Morales for analyzing, and processing SIA samples. The community volunteers who allowed us to catch birds at their homes! VJA is supported by the NSF Postdoctoral Research Fellowships in Biology Program under Grant No. 2305367.

References: 1. Greig, E. I., Wood, E. M., & Bonter, D. N. Winter range expansion of a hummingbird is associated with urbanization and supplementary feeding. *Proc Biol Sci* 284, 20170256 (2017). 2. Meadows, M. G., Roudybush, T. E. & McGraw, K. J. Dietary protein level affects iridescent coloration in Anna's hummingbirds, *Calypte anna*. *J Exp Biol* 215, 2742–2750 (2012). 3. Vaz, S. et al. Global meta-analysis of urbanization stressors on insect abundance, richness, and traits. *Science of The Total Environment* 903, 165967 (2023). 4. Hardesty, J. Using nitrogen-15 to examine protein sources in hummingbird diets: Uso de nitrógeno-15 para examinar las fuentes de proteína en las dietas de los colibríes. *Ornitología Colombiana* 19–28 (2009) doi:10.59517/oc.e183. 5. Wolf, N., Smeltz, T. S., Cook, C. & Martínez del Río, C. Using stable isotopes in hummingbird breath to estimate reliance on supplemental feeders. *Ecology and Evolution* 13, e9799 (2023).